

TMS320DM270 Issue List

Rev.1.2

as of Mar/24/2003

No.	Module	Title	Classification	Description
1	SDRC	SDRAM Performance	H/W issue	Change the target performance from 120MHz to 100MHz
2	SDRC (USB)	DMA Deadlock When ARM > SDRAM	H/W issue	When ARM is operated at higher frequency than SDRAM, the USB DMA deadlock might happen. (Same modification method as Item 6, 8 and 9)
3	USB	DMA Abnormally Terminated in Multi-Packet Mode	H/W issue	If a USB transfer for the endpoints other than the DMA target is issued from the USB host during DMA running and this transfer is completed with ACK, the DMA is abnormally terminated at ACK handshake. (Same modification method as Item 4)
4	USB	Data Lost In DMA FIFO Read/Write Mode	H/W issue	Only affected for the DMA FIFO read/write mode. If DMA runs while the USB transfer is on-going, data corruption will happen. (Same modification method as Item 3)
5	USB	IN Fifo Data Corruption When ARM >= 72MHz	H/W issue	When ARM operates at 72MHz or higher frequency, IN transmission would be abnormally corrupted.
6	SDRC (HPIB)	Restricted frequency in DMA	S/W limit.	Restricted frequency relation between ARM clock and SDRAM one at using DMA. HPIB : ARM < SDRAM x 2 (Same modification method as Item 2, 8 and 9)
7	IMG	Restricted frequency in DMA	S/W limit.	Restricted frequency relation between SDRAM clock and AXL one at using DMA. AXL-SDRAM : AXL = SDRAM x N (N=2,3,4,...)
8	EMIF	Restricted frequency in DMA	S/W limit.	Restricted frequency relation as between SDRAM clock and ARM one at using DMA. SDRAM-SDRAM : ARM < SDRAM (Same modification method as Item 2, 6 and 9)
9	EMIF	Restricted frequency in DMA	S/W limit.	Restricted frequency relation as between SDRAM clock and ARM one at using DMA. SDRAM-EMIF : ARM < SDRAM x 3 (Same modification method as Item 2, 6 and 8)
10	CLKC	PLLA outputs in ext. host mode	H/W issue	PLLA outputs the low frequency (120MHz) at boot from External Host IF. Need the detail in characterization.
11	SDRC	SDRC priority register initial configuration	S/W requirement	Priorities of SDPRTY9 and SDPRTY10 are needed to swap. SDRAM refresh can't be occurred, unless change the register settings.
12	SIF	Assertion timing lag of XMIT	H/W issue	Transfer status bit (XMIT) can't indicate correctly during transferring.
13	DSP Subchip	API code	S/W issue	API code bug. (MemoryRdFast_NonWait, MemoryRdFast_Wait, MemoryWrFast_NonWait, MemoryWrFast_Wait)
14	HPIB	Unstable HPI access cycle	H/W improvement	HPI access in QC min condition might be 1 cycle less than QC max. This makes DSP program based TDL generation more difficult. It can be fixed by only inverting HPIB DSP clock. No affect on SCAN clock.
15	USB	USB Compliance	H/W issue	Invalid response to SET/CLEAR_FEATURE request. If the USB host issued a SET/CLEAR feature request with an illegal feature selector value, the UDC controller incorrectly responded to that request with an ACK handshake. It violated the USB 1.1 protocol since the UDC controller responded with an ACK rather than a STALL handshake to the USB host.
16	EHIF	Unstable external Host	H/W improvement	In external host mode, noise on EM_CE/EM_WE/EM_OE pin is propagated to clock port of first F/F. Therefore, read/write access occasionally doesn't work.
17	BUSC / EHIF	Revision Register	Spec. change	Update revision register to be read "11h" from ARM and external host.
18	SDRC	Mobile command	H/W improvement	Support extended mode for mobile SDRAM.
19	VENC	DAC powerdown	H/W improvement	Change initial value of powerdown configuration register (VID01(0x30800) DAPD (bit5)) to "1" by default.
20	USB	UDC Bug	H/W issue	A packet error may occurs by misdetecting SE0 state when SE0 state which has less than 14ns happend along of the skew between USB_DP and USB_DM.
21	SDRC	Refresh request	H/W improvement	When SDRAM refresh request is occurred, the priority of SET signal is higher than RESET in SR logic. For failsafe operation, RESET will be prioritized over SET signal.
22	CLKC	enable DSP clock	Test spec. change	When BurnIn test, DSP can not make reset at bist mode because clk is not active.
23	VENC	CASIO LCD I/F	H/W issue	CASIO LCD I/F does not work correctly. Simultaneous display with analog composite (NTSC/PAL) could not be achieved.
24	VENC	external sync malfunction	H/W issue	The external sync function is added from DM270. But the mux insert point for external sync select was inappropriate. The external sync function does not work correctly.
25	SDRAM I/O	SDRAM data bus pull-down	Test Spec. change	Enable pull down on SDRAM data bus when a register set to "1". This function is related to test. (SDR_DQx=pulldn enable, SDR_xxx=normal function)
26	SDRAM clock	SDRAM clock activation	Spec. change	Disable SDRAM clock by default. Will change default polarity of clock enable register.
27	VENC	External sync latch phase	H/W issue	In spite that external sync has to be latched in synchronized with VENC internal 13.5MHz clock (sclk), this was not be done.
28	VENC	REC601 invalid data mask	H/W issue	In REC601 without sav/eav insertion, the horizontal data mask assertion was misaligned to the counter value which determines YCbCr output order. The mask signals are synchronized with 13.5MHz and the counter is sync with 27MHz, but there is one 27MHz phase lag between them.
29	PRV	wait request in CCD input mode	H/W issue	In spite of CCD input, PRV module made wait signal active when a half of FIFO was filled. It wasted capacity of FIFO, and the limit made warped image view when frequent access to SDRAM. Then the wait signal in CCD input was disables .
30	OSD	Across noise on video windows	H/W issue	A horizontal line coinciding with the bottom or top line of video window 1 is seen across video window 0. This across noise comes up when the vertical filter is on.
31	OSD	Band of irregular filter	H/W issue	There is zonate areas which is filtering with wrong filter factor. As the results, filtered or non filtered areas are seen in adjacent video windows 1.